

Refill Alert: LPG Cylinder Monitoring and Alert Notification System

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Abstract—Liquefied Petroleum Gas (LPG) is widely used for household and industrial applications, where continuous monitoring and safety are critical. This paper presents a LPG monitoring system that measures, analyzes, and communicates the cylinder status in real time. The proposed system uses a load cell sensor with an HX711 amplifier to measure the cylinder weight and calculate the remaining gas level. An Arduino Uno microcontroller processes the sensor data and compares it with a predefined threshold value. When the gas level falls below the threshold, the system activates a buzzer and sends an SMS alert to the user through a GSM module. Experimental results show that the system detects low gas conditions quickly and sends notifications within approximately five seconds. The use of GSM communication enables the system to operate effectively without requiring internet connectivity, making it suitable for both urban and rural environments. Unlike traditional methods that depend on manual inspection, such as tapping or lifting the cylinder, the proposed approach provides accurate and reliable monitoring through weight measurement. Overall, the proposed solution offers a low-cost, efficient, and user-friendly approach for real-time LPG monitoring and safety management.

Index Terms—LPG Cylinder Monitoring, Load Cell Sensor, GSM-Based Alert System, Arduino Uno

I. INTRODUCTION

A. Motivation towards the problem

In present days the main problem arises from the common difficulty faced by every household in identifying the remaining gas in an LPG cylinder. Shaking and pouring hot water on the cylinder are very unsafe practices that are still at large and pose safety hazards, leading to unexpected depletion. Many available smart systems make use of mobile applications or an Internet network, which is unpredictable, especially in rural areas. Therefore, the need exists for a simple, effective, and low-cost solution that can function without the

need for the Internet. This research aims to develop a reliable method for accurately monitoring the remaining Liquefied Petroleum Gas (LPG) level in cylinders and providing information to users before the gas fully runs out.

B. Limitations with existing systems

Present existing systems have some limitations, which are impractical and unreliable for everyday use. [1][2][3] Most of the existing systems are heavily dependent on Internet connectivity and mobile applications, rendering them unsuitable in rural and semiurban areas where good network access is poor. Their implementations are often very expensive due to the involvement of network and cloud-based components, increasing their maintenance expenses and reducing affordability. Most of the existing systems lack offline working and are not independent of the Internet, making them less usable.

C. Proposed Solution and Contributions

To overcome those limitations an efficient and cost-effective LPG monitoring system has, therefore, been developed for addressing these problems. The system consists of a load cell, HX711 amplifier, microcontroller, and a GSM module that measures the weight of the cylinder continuously, process the information, and send SMS alerts directly to the user in cases of low levels of gas. This eliminates guesswork, reduces human effort, and enables the user to plan timely refills without access to the Internet. It also reduces risk factors since the cylinder is handled less. [4] The design works well for homes, commercial kitchens, and rural areas. It also has the future potential to support IoT apps, and cloud storage, and predictive analytics for

automated refilling, making it a practical step towards better LPG management. The novelty of the proposed LPG monitoring system lies in its cost-effective and internet-independent design.

Key Contributions:

- Automatic Alert System: It automatically sends SMS notifications within seconds when the gas level falls below a predefined threshold value.
- GSM Communication: Communication and alerts even in areas that do not have an internet connection.
- Versatile approach: Can be used in both domestic and commercial LPG applications because of its standalone GSM-based architecture.

II. BACKGROUND RESEARCH

To make it simpler to monitor LPG cylinders and increase user safety, numerous studies and projects have been initiated. In [5], a system was proposed that calculates the residual quantity of the gas and percentage consumed using a load cell for measurement of weight in real time. In case of the detection of a gas leak, it automatically cuts off the regulator and buzzes to notify the user. It also sends a notification to their app. It provides an in-app one-click refill booking feature and is designed conveniently as a cylinder stand. Similarly, [6] presents a system for the measurement of LPG level inside the cylinder, with automatic shut-off of the regulator in case of a gas leak for safety. The work reported in [7] presents a user-friendly interface for real-time monitoring and control of gas usage via a mobile app to get instant alerts, check gas levels, and operate the gas valve for better safety and convenience. In [8], a project was developed, that project on LPG-level monitoring and its leakage detection using advanced sensors that are quick and more accurate in detecting leakage. Ab Aziz et al.[9] proposed an IoT-based Cooking Gas Monitoring and Alert System that tracks LPG levels and detects leakage in real time. The setup includes an MQ-6 gas sensor, load cell with HX711, Node MCU ESP8266, buzzer, LEDs, and a 16×2 LCD for local monitoring. Cylinder weight and surrounding air are continuously analyzed to identify low gas (below 10or leakage conditions. Alerts are instantly delivered to users via the Blynk mobile app along with buzzer and LED indications. Experimental results show reliable performance, with future improvements focusing on enhanced durability and automated cylinder capacity configuration. And Bairagi et al. [10]proposed a WSN-based LPG monitoring and leakage detection alert automation system that includes gas level measurement, and automated re-filling booking. The system implements an MQ-4 sensor, a load cell with the HX711 amplifier, GSM module, WiFi module ESP8266, LCD, and Arduino UNO as the main controller. The cylinder's weight is continuously monitored by the load cell, which triggers an SMS-based auto booking when

the minimum threshold is reached. The MQ-4 sensor detects leakage. and immediately activates a buzzer and safety alert notifications. In communication, the distance does not impact data transmission; operation occurs wirelessly, hence guaranteeing installation flexibility. Experimental tests validated that it provides real-time detection accuracy, timely messaging, and autonomous booking functionality. This will help the users avoid running out of gasoline at the last minute. Furthermore, [11][12] has discussed the development of the system that will trigger alerts to the user through a mobile application and buzzer when the level of gas goes below the threshold level. These studies collectively highlight the growing focus on automation, safety, and convenience in LPG monitoring systems. These studies collectively highlight the growing focus on automation, safety, and convenience in LPG monitoring systems, also demonstrate the need for smarter, real-time monitoring solutions to ensure efficient gas usage and timely alerts.

III. METHODOLOGY

The LPG monitoring system proposed in this paper operates by continuously measuring weight, processing signals, and communicating wirelessly. A load cell, positioned at the bottom of the LPG cylinder, measures the continuous change of weight when the gas is consumed. A load cell, located at the bottom of the LPG cylinder, monitors the weight change continuously as the gas is used. The load cell generates the small analog signals that are carried through to the HX711 amplifier module to produce an accurate digital output. These readings are then sent to the Arduino Uno microcontroller, which calculates the remaining gas level based on the calibration values set for full and empty cylinders. In the event of a low range of gas where the gas level falls below a threshold value, the system triggers a buzzer and sends an SMS to a registered mobile number detailing a low level of gas, as the level falls below a threshold level around which a full cylinder is calibrated. The monitoring system provides a means of tracking the use of LPG in each home, without the user needed to monitor, with convenience, safety, and timely refilling as depicted in Fig. 1 and Fig. 2 explains how the system works.

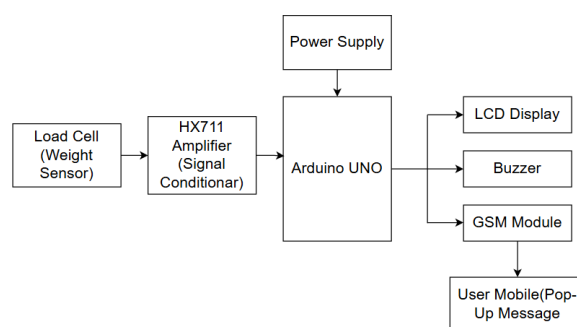


Fig. 1. Block diagram of the Proposed system

The objective of the monitoring module is to detect when the LPG level falls to a level that is considered critical, such as 20detected level, the microcontroller triggers the GSM communication unit to send an SMS message to the registered mobile number of the user so that he can plan for its refill in time. A buzzer also buzzes to alert audibly if somebody is nearby for better safety and awareness. In some designs, an LCD display can show the present weight of the cylinder for easy local monitoring. Since this system is developed, it will help to avoid delays in scheduling a refill with a new cylinder and manage continuous serving in both domestic and commercial kitchens. It is of a compact, low-cost, and reliable design to be installed anywhere. This system will avoid delays in scheduling a new cylinder and provides uninterrupted use in homes and commercial kitchens [14]. The approach focuses on a rigid, low-cost, and reliable structure that can be set up anywhere. In the future, the system may also include IoT connectivity, the mobile app dashboards, and the cloud storage to examine gas consumption patterns and enable remote management. Thus, the approach offers a convenient and safety-focused way of managing the LPG effectively.

IV. ALGORITHM

Algorithm LPG Cylinder Monitoring and Alert Notification System

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1: Input: Weight of the LPG cylinder
2: Initialize Load Cell, GSM Module, LCD Display and Buzzer
3: Set predefined values: Threshold
4: while true do
5:   weight  $\leftarrow$  Loadcell.read Weight()
6:   if weight  $\leq$  Threshold then
7:     Display GAS LEVEL LOW
8:     Activate Buzzer
9:     (active_signal)  $\leftarrow$  GSM_Module.Activate()
10:    Prepare SMS message as GAS LEVEL LOW
11:    Send SMS to User via GSM Module
12:   else
13:     Display NORMAL
14:     Continue monitoring
15:   end if
16:   Wait for short delay (1–2 seconds)
17: end while
18: Output: Low gas notification to the user

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V. COMPONENTS AND SYSTEM BLOCKS

Power Supply: The power supply provides the voltage that the system requires. It changes the standard 230V AC voltage to DC voltage, so that all components receive the power that is required for proper and

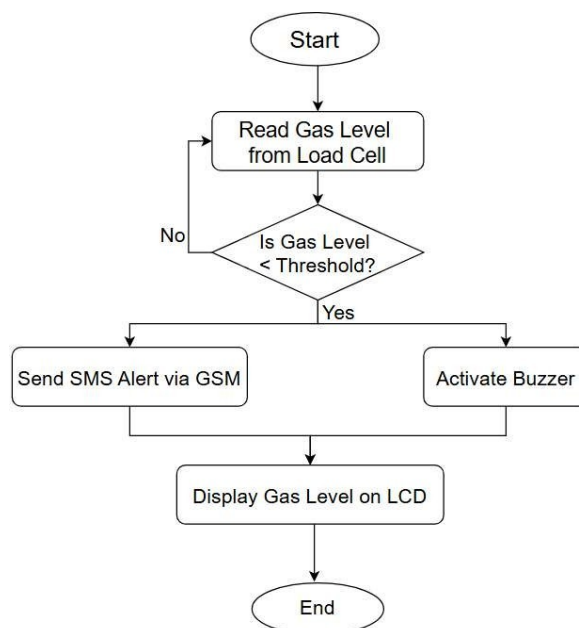


Fig. 2. Workflow of the system

safe operation. The Arduino microcontroller consumes approximately 40–50 mA, the HX711 load cell amplifier consumes less than 10 mA, and the GSM module consumes 300–500 mA during SMS transmission. During normal operation (without GSM transmission), the system consumes low current, making it energy-efficient.

Arduino Uno Module: The Arduino Uno acts as the central control unit of the system. The Arduino board collects the input data from sensors, processes the readings, and controls output devices[15]. The Arduino Uno was selected as the unit is easy/straightforward to use, easy to program, low cost, and provides the best supportive community.

Load Cell and HX711 Amplifier: The load cell measures the weight of the LPG cylinder, then the amplifier HX711 converts it from an analog signal into a digital signal to make it understandable by the Arduino.

LCD Display: An LCD display, acts as the interface to the user and displays the instantaneous indication of the level of LPG in the cylinder along with system alerts. This gives the user the ability to look at the LCD and immediately comprehend the state of the cylinder.

GSM Module: In order to keep the user informed even when they are not at home, the GSM module is used for communication purposes that works without internet, and sends an SMS alert.

Buzzer: It is used to indicate the real-time user through sound and alert the situation, and after sending the notification buzzer shuts off.

With home automation, consumers may monitor and operate gadgets from a distance without having to be there. When if the primary user is illiterate, their family members or children can easily handle the

alert notifications send by the GSM Module. These improvements in mobile technology and the Internet of Things (IoT) enable better convenience in making LPG cylinder bookings.

VI. SYSTEM IMPLEMENTATION

The proposed system mounts a load cell sensor underneath the gas cylinder for continuous weight measurement. The analog signals from the load cell, after conversion to digital readings by the HX711 module, are transferred to the Arduino microcontroller. The controller interprets the readings to calculate the remaining LPG level with calibrated values for full and empty cylinders. When it detects the level of gas below the threshold value, an alerting SMS is sent to the user's mobile phone through the GSM module, while a buzzer is triggered as an instant warning when the level falls below a certain threshold. For convenience, the current gas level is shown on the optionally provided LCD display.

A. Hardware Setup

The hardware setup as shown in Fig. 3 consists of a load cell placed under the LPG cylinder, which measures weight changes. To convert weak analog signals to digital data, the load cell is connected to the HX711 amplifier, while the Arduino UNO will process these signals to find the remaining gas quantity. The GSM module connects to the Arduino for alerting the user via SMS. A 5V adapter provides regulated power supply to the entire system. Other peripherals are lcd and buzzers that show system activity and alert conditions. Proper wiring and sensor mounting help in avoiding noise in data collection and dependability of operation.

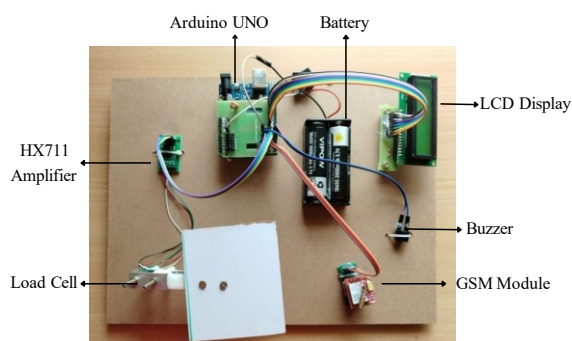


Fig. 3. Hardware setup

VII. RESULTS AND DISCUSSION

Gas cylinders of various weights were used to test the accuracy and alert response of the system when in use. We observed the readings for comparison with the actual weights and displayed them in a graph, as shown in Fig. 8. The Alert timing and reliability of

the GSM module were tested. The data show that the system was able to detect gas levels within less than a set margin. It takes 4 to 5 seconds to detect and compare with the threshold value and then the alerts are sent in 2 to 5 seconds. For the case of low signals to the SIM800L, it takes 7 to 10 seconds to send the alert notification. The power consumption was within safe limits, thus allowing continuous operation. The GSM communication delay depends on signal strength, network congestion, SMS routing mechanisms, and service provider latency. While we are doing tests, an average delay of 5-15 seconds was observed between threshold detection and SMS delivery. The delay variation is caused by queuing in the cellular network and the signal propagation delays. The system performs exceptionally well for the purpose of monitoring the real-time LPG levels, the real-time example is shown in Fig. 4. It sent alerts on time and ensured safety at home.

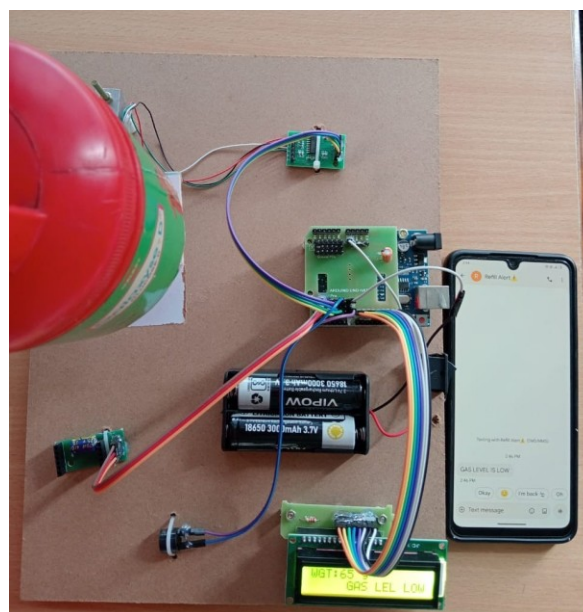


Fig. 4. Real-time Output observed on Mobile and LCD display

A. Test Cases

The list of sample text cases that follows was taken into account:

Test Case-1: Normal Gas Level

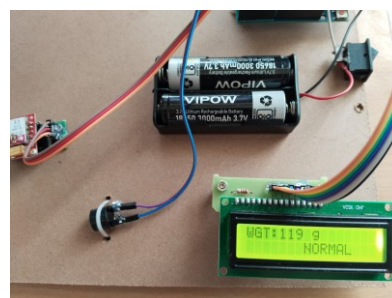


Fig. 5. Normal Gas level in LCD Display

When the detected LPG weight exceeds 50 g, the system concludes that the gas level is normal. In five seconds, the weight is processed, and the display is updated accordingly. In this case, no SMS alert is sent because the gas level is sufficient as shown in fig.5 and doesn't require user intervention.

Test Case-2: Threshold Level

When the detected weight is exactly 50 g, the system recognizes that the gas level has dropped below the minimum threshold as shown in Fig. 6. It detects this condition in five seconds and sends an SMS to the registered user right away. Usually arriving in two to three seconds, the alert ensures timely notification for refill scheduling.

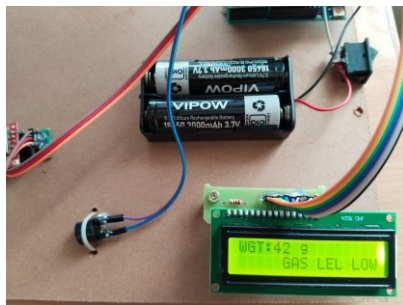


Fig. 6. Below Threshold Gas level in LCD Display

Test Case-3: Low Gas in Weak Network Area

The system detects the low gas level in 5 seconds even when the detected weight falls below 50 g in a weak network area. However, signal problems may cause the SMS alert delivery to take longer—usually 7–10 seconds. The system successfully notifies the user before the cylinder runs completely empty in spite of this delay.

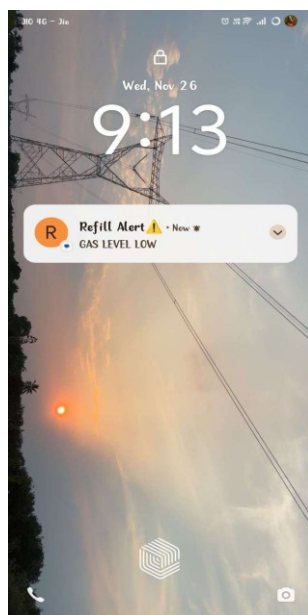


Fig. 7. Notification in Mobile

Table II shows the system performance in different gas level conditions. When the gas level is normal, only the display updates and no SMS is sent. At the threshold level, the system detects the condition within 5 seconds and sends the SMS alert within 2–3 seconds. In weak network areas, although the detection remains fast, the SMS delivery time increases to 7–10 seconds. This confirms that the system works accurately and efficiently, with minor delays only in low-signal conditions.

TABLE I
GAS LEVEL DETECTION AND SMS ALERT RESPONSE

Condition	Detected Weight	Detection Time	SMS Send-ing Time
Normal Gas Level	> 50 g	< 5 sec	No SMS
Threshold Reached	50 g	< 5 sec	2–3 sec
Weak Network Area	< 50 g	< 5 sec	7–10 sec

The accuracy of the proposed system is demonstrated by its consistent and correct detection of gas level conditions. The system accurately distinguishes between normal and threshold gas levels, triggering alerts only when the predefined threshold is exceeded. Detection time remains within 7 seconds across all test conditions, indicating reliable sensor performance and precise threshold comparison. These results confirm that the system provides accurate and dependable monitoring.

$$\text{Accuracy (\%)} = \left(1 - \frac{|\text{Actual} - \text{Measured}|}{\text{Actual}}\right) \times 100$$

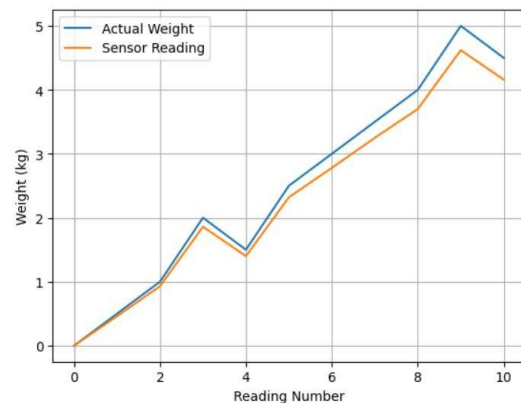


Fig. 8. Actual Vs Measured Graph

VIII. CONCLUSION

This research has considered many factors. The proposed LPG monitoring system is one of the safest and most cost-effective ways to monitor the level of gas. The system is made up of a load cell, Arduino, a GSM module, and an HX711 amplifier. The proposed system will automatically send an SMS alert when

the gas level falls below the predetermined level. This will make the process of refilling the gas automatic and will not require manual monitoring. In addition, the refilling process will occur at the right time. Since internet connectivity is not a requirement for the functionality of this system, this project can be done in both rural and urban settings. To enhance a more complex model of LPG management, future projects can incorporate Internet of Things connectivity, cloud monitoring, and mobile application functionality.

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